

Sentiment Analysis of Sephora Product Reviews

MGTA415 - Analyzing Unstructured Data
Professor Jingbo Shang
Tweety Chang¹, Yoonjeong Park¹, Emma Wu¹

¹ University of California, San Diego

Abstract—Consumer sentiment analysis helps understand brand perception and enhance customer engagement in the cosmetics industry. This study analyzes Sephora product reviews using natural language processing (NLP) techniques to determine the most effective machine learning approach for sentiment classification while addressing dataset imbalances. We conduct exploratory data analysis (EDA), apply text preprocessing methods, and evaluate three text vectorization techniques—Count Vectorization, TF-IDF, and Word2Vec—combined with logistic regression. Hyperparameter tuning and cross-validation are used to optimize performance. Results show TF-IDF with logistic regression achieves the best balance across sentiment classes. These insights help brands leverage customer feedback for data-driven decisions and product improvements.

I. INTRODUCTION

In today’s digital landscape, online reviews play a crucial role in shaping consumer purchasing decisions. Sephora, a leading beauty retailer, receives thousands of customer reviews on its products, providing valuable insights into customer satisfaction and preferences. However, manually analyzing such a large volume of reviews is inefficient and impractical.

This is where sentiment analysis becomes essential. Sentiment analysis is a method of using natural language processing (NLP) and machine learning to determine whether customer reviews express positive, neutral, or negative sentiments. Understanding customer sentiment helps businesses identify strengths and areas for improvement, refine marketing strategies, and enhance product offerings. For Sephora, this analysis can provide deeper insights into customer experiences, helping the company improve its product recommendations and customer engagement. The code used in this project is publicly available at github.com/yop018/MGTA415_FinalProject.

II. DATASET

Our dataset consists of nearly 50,000 product reviews, including customer ratings, recommendations, and written feedback. Many reviews contain unstructured text, making it essential to preprocess and transform the data into a format suitable for machine learning models.

The distribution of ratings in the dataset is highly imbalanced, with the majority of reviews receiving a rating of 5, while ratings of 1, 2, 3, and 4 are significantly underrepresented. This imbalance suggests a strong positive bias, where most customers tend to leave highly favorable reviews, possibly due to customer satisfaction bias, social desirability effects, or promotional incentives influencing review behavior. As a result, the sentiment distribution becomes skewed, making it difficult to achieve a balanced classification of customer opinions. In sentiment analysis and machine learning classification, such an imbalance can lead to biased predictions, where the model heavily favors positive sentiment while struggling to correctly classify neutral and negative sentiments. This results in overfitting to the dominant class (rating 5), causing the model to misinterpret critical reviews as positive, leading to poor classification performance and misleading sentiment insights.

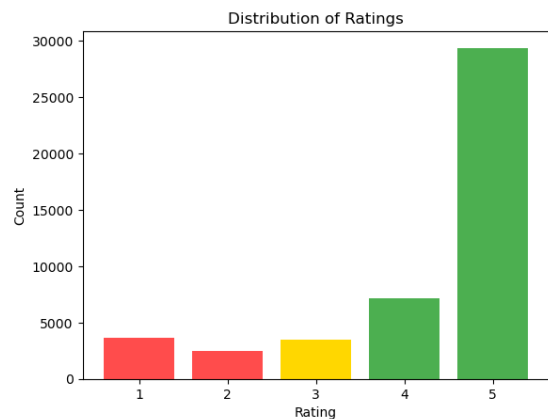


Fig. 1. Dataset Distribution of Ratings

Similarly, the distribution of `is_recommended` in the dataset is highly imbalanced, with a significantly higher number of reviews marked as recommended (1.0) compared to those labeled as not recommended (0.0). This imbalance suggests that most customers who leave reviews tend to recommend the product, whereas a much smaller proportion explicitly choose not to recommend it. Such a skewed distribution can introduce bias into sentiment classification models, leading them to disproportionately predict recommended outcomes while failing to accurately classify not recommended cases. This issue is particularly concerning in machine learning-based sentiment analysis, where class imbalance can lead to overfitting to the majority class and reduced sensitivity to minority cases. As a result, the model may struggle to distinguish between genuinely dissatisfied customers and those who still recommend the product despite expressing concerns.

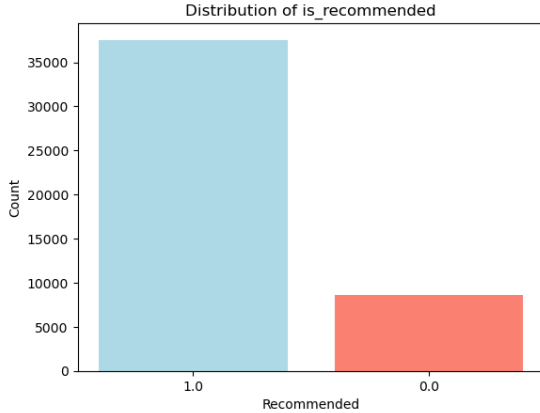


Fig. 2. Dataset Distribution of `is_recommended`

To address the dataset imbalance, class weighting was applied in the logistic regression model by setting `class_weight='balanced'`. This technique automatically adjusts the weights assigned to each class, ensuring that the model does not favor the majority class (e.g., highly rated or recommended reviews) while underrepresenting the minority classes. By incorporating this approach, the model is encouraged to make fairer classifications across all sentiment categories, improving its ability to correctly identify both positive and negative sentiments despite the dataset’s skewed distribution.

The dataset consists of a total of 49,977 reviews, from which 46,160 cleaned reviews were utilized for analysis. The dataset was partitioned into training and test sets using a stratified random sampling approach, maintaining a 75% to 25% ratio. Consequently, the training set comprises 36,928 samples, while the test set contains 9,232 samples. The structured data distribution

Column Name	Description
rating	Customer’s product rating (1–5)
is_recommended	If the reviewer recommends the product (1: Yes, 0: No)
full_review	Combined text from <i>review_title</i> and <i>review_text</i> , after cleaning and merging
sentiment	Sentiment category (positive, neutral, negative)
preprocessed_review	Processed text (tokenized, lemmatized, stop-words removed)

TABLE I
TABLE I: DATASET COLUMNS AND DESCRIPTIONS

ensures balanced representation for model evaluation and enhances the reliability of sentiment classification.

Dataset	Information
Total review	49,977
Cleaned Reviews Used in Analysis	46,160
Training Data Size	36,928
Test Data Size	9,232

TABLE II
TABLE II: DATASET INFORMATION

III. LITERATURE

Sentiment analysis, a crucial field in natural language processing (NLP), is widely used to classify customer feedback into sentiment categories such as positive, neutral, and negative. With the growing reliance on e-commerce reviews, businesses utilize sentiment analysis to interpret consumer preferences, enhance product offerings, and improve marketing strategies [1].

The sentiment classification process involves several key components, including data preprocessing, feature extraction, model selection, and evaluation metrics. Preprocessing steps such as tokenization, stop-word removal, lemmatization, and vectorization ensure that textual data is standardized for machine learning models [2]. The study employs three feature extraction techniques—Count Vectorization, TF-IDF, and Word2Vec embeddings—to transform text into numerical formats suitable for classification.

Previous research highlights the strengths and limitations of these techniques. Count Vectorization, which represents text as word frequency counts, is effective for simple text classification but lacks contextual understanding [3]. TF-IDF improves upon Count Vectorization by reducing the influence of common words while emphasizing distinguishing terms, making it more effective for sentiment classification tasks. Word2Vec, a deep learning-based approach, captures semantic relationships between words, though its performance depends on dataset size and complexity [1].

In sentiment analysis, model selection and training are pivotal to achieving high classification accuracy. This

study employs logistic regression, a widely used text classification model due to its efficiency and interpretability [2]. However, class imbalance remains a significant challenge, as the majority of reviews are positive. Techniques such as SMOTE (Synthetic Minority Over-sampling Technique) and class weighting are utilized to address this issue [3]. Sarcasm and implicit sentiment pose additional challenges, where contextual meaning can drastically affect classification outcomes. Advanced deep learning models, such as BERT and LSTMs, have been shown to capture these contextual nuances more effectively [1].

Evaluation metrics such as accuracy and macro-F1 score are crucial in sentiment analysis.[2] emphasizes that while accuracy measures overall correctness, macro-F1 score provides a balanced evaluation by considering precision and recall across all sentiment categories . The study’s findings indicate that Count Vectorization achieves the highest accuracy (84.3%), whereas TF-IDF outperforms other methods in macro-F1 score (0.680), making it the most balanced classifier. Word2Vec, despite its semantic capabilities, performs poorly due to dataset limitations.

IV. METHODOLOGY

This study employs sentiment analysis to classify Sephora product reviews into positive, neutral, and negative sentiments using natural language processing (NLP) and machine learning techniques. The methodology is structured into four main phases: data collection and preprocessing, feature engineering, model selection and training, and evaluation metrics.

A. Data Collection and Preprocessing

The dataset consists of 49,977 customer reviews, including numerical ratings, written reviews, and recommendation indicators. To ensure data quality and consistency, the following preprocessing steps were undertaken.

a.1) Handling Missing Values

The dataset contained missing values in several fields, notably *is_recommended*, *review_text*, and *review_title*. To maintain data integrity, records with missing *is_recommended* values were removed. Meanwhile, missing values in *review_text* and *review_title* were replaced with empty strings, and both fields were combined into a new column, *full_review*, representing the complete customer review.

a.2) Sentiment Labeling

Sentiment labels were created using the *rating* column, where ratings of 4 or 5 were classified as *positive*, a

rating of 3 as *neutral*, and ratings of 1 or 2 as *negative*. This labeling method provides a structured approach to categorizing customer feedback based on numerical scores.

a.3) Text Preprocessing

To prepare textual data for machine learning models, various NLP techniques were applied. Lowercasing was used to standardize text by converting all words to lowercase, while punctuation and special characters were removed to retain only alphabetical characters. Tokenization was performed (e.g., using NLTK) to split text into individual words, followed by stopwords removal to eliminate common words that do not contribute to sentiment. Lemmatization then converted words to their base form (e.g., “running” → “run”). Finally, the processed text was stored in a new column, *preprocessed_review*.

B. Feature Engineering

To convert textual data into a numerical format for machine learning, three text vectorization techniques were used: Count Vectorization, which converts text into a matrix of token counts; TF-IDF (Term Frequency-Inverse Document Frequency), which assigns importance to words based on their relative frequency; and Word2Vec Embeddings, which generates word representations by learning word relationships from the dataset. Each of these vectorization methods was combined with logistic regression to assess their effectiveness in sentiment classification.

C. Model Selection and Training

The dataset was split into 80% training and 20% testing to train and evaluate the models. To ensure optimal performance, the following procedures were followed:

c.1) Model Selection

Logistic regression was chosen as the classification model for its effectiveness in text classification tasks, and the impact of different vectorization techniques (Count Vectorization, TF-IDF, and Word2Vec) was analyzed to evaluate their performance.

c.2) Hyperparameter Tuning

GridSearchCV was applied to fine-tune hyperparameters and identify the best-performing model configuration, while Stratified K-Fold Cross-Validation was used to ensure robust performance across different data splits and effectively handle class imbalances.

D. Evaluation Metrics

To evaluate model performance comprehensively, accuracy was used to measure the overall correctness of predictions, providing a general assessment of how well the model classified sentiments. However, since accuracy alone can be misleading in imbalanced datasets, the macro-F1 score was also utilized. This metric balances precision and recall across all sentiment categories, ensuring that minority classes are not overshadowed by dominant ones. By incorporating both accuracy and macro-F1 score, the evaluation framework provided a more reliable measure of the model’s effectiveness in sentiment classification.

V. EXPERIMENTS

The experimental design for this study aims to evaluate the effectiveness of different text vectorization techniques in sentiment classification of Sephora product reviews. We conducted controlled experiments using three feature extraction methods—Count Vectorization, TF-IDF, and Word2Vec—combined with logistic regression. The performance of each model was assessed based on accuracy and macro-F1 score to ensure balanced evaluation across all sentiment classes. Additionally, hyperparameter tuning was applied to optimize model performance.

A. Experimental Setup

The dataset was divided into training (80%) and testing (20%) sets using stratified sampling to maintain sentiment class distributions. The training set was used for model learning and hyperparameter optimization, while the test set was reserved for final performance evaluation.

a.1) Preprocessing Pipeline

To prepare the text data for modeling, several preprocessing steps were applied to enhance the quality and consistency of the input. Tokenization was performed to break down review text into individual words, facilitating further processing. Stopword removal eliminated common words that contribute little to sentiment analysis, reducing noise in the dataset. Lemmatization was then applied to convert words into their root forms, improving generalization and reducing redundancy. Finally, text vectorization techniques, including Count Vectorization, TF-IDF, and Word2Vec, were used to transform textual data into numerical representations suitable for machine learning models, ensuring effective sentiment classification.

B. Experimental Setup

To transform unstructured text into a suitable format for machine learning, we employ three widely used feature extraction techniques: Count Vectorization, TF-IDF, and Word2Vec. Each method has unique characteristics and hyperparameters that influence classification performance. The following subsections describe these approaches in detail

b.1) Count Vectorization (Bag-of-Words Approach)

Count Vectorization represents text as a sparse matrix, where each column corresponds to a word, and the values indicate word frequency. While this method effectively captures basic term frequency information, it does not account for word importance or contextual relationships, making it less sophisticated than other vectorization techniques like TF-IDF or Word2Vec. To optimize its performance, hyperparameter tuning was applied to key settings. The `max_features` parameter controlled the number of top words retained in the matrix, with values tested at 3,000, 5,000, and 10,000. Additionally, the `ngram_range` parameter was adjusted to compare the impact of unigrams (1,1) versus bigrams (1,2) in capturing meaningful textual patterns. These optimizations aimed to improve Count Vectorization’s ability to represent text effectively for sentiment classification.

b.2) TF-IDF (Term Frequency-Inverse Document Frequency)

TF-IDF enhances Count Vectorization by weighting words based on their importance within a document relative to their overall frequency in the corpus. This approach reduces the influence of common words while emphasizing distinguishing terms, making it more effective for sentiment analysis. To optimize TF-IDF performance, hyperparameter tuning was conducted on key parameters. The `max_features` setting controlled the number of top words retained in the model, with values tested at 3,000, 5,000, and 8,000. Additionally, the `ngram_range` parameter was adjusted to compare the effectiveness of unigrams (1,1) versus bigrams (1,2) in capturing meaningful patterns in the text. These optimizations ensured that the TF-IDF vectorizer effectively captured sentiment-relevant features for classification.

b.3) Word2Vec (Word Embeddings)

Unlike Count Vectorization and TF-IDF, which rely on term frequency, Word2Vec learns word representations by training a neural network to map words into a continuous vector space. This approach captures semantic relationships between words, allowing for a deeper understanding of word meaning beyond simple frequency-based

methods. To optimize Word2Vec’s performance, several hyperparameter tuning options were explored, including `vector_size`, which determines the dimensionality of word embeddings (values tested: 50, 100), and window size, which defines the context window for word relationships (values tested: 3, 5). Additionally, `min_count` was adjusted to set the minimum word occurrence required for inclusion in the model (values tested: 1, 2). Lastly, the `sg` parameter controlled the training algorithm, comparing Continuous Bag of Words (CBOW) and Skip-gram approaches (values tested: 0, 1), ensuring an optimal configuration for learning word representations.

VI. RESULTS

To evaluate the effectiveness of different text vectorization techniques, we assessed the performance of Count Vectorization, TF-IDF, and Word2Vec, each paired with logistic regression. The models were evaluated using accuracy and macro-F1 score, ensuring a comprehensive analysis of overall correctness and balanced performance across sentiment classes.

Method	Best Accuracy	Best Macro-F1 Score
Count Vectorization + LR	84.3%	0.674
TF-IDF + LR	84.1%	0.680
Word2Vec + LR	79.3%	0.639

TABLE III
PERFORMANCE COMPARISON

Count Vectorization + Logistic Regression achieved the highest accuracy (84.3%), indicating that it provided the most overall correct classifications. However, due to the dataset’s imbalance, this approach may be biased toward dominant sentiment classes, such as positive sentiment.

In contrast, TF-IDF + Logistic Regression achieved the highest macro-F1 score (0.680), making it the most balanced model in terms of performance across all sentiment categories. This suggests that TF-IDF is more effective at distinguishing between positive, neutral, and negative sentiments compared to Count Vectorization, which may favor the majority class. On the other hand, Word2Vec + Logistic Regression had the lowest performance, with an accuracy of 79.3% and a macro-F1 score of 0.639. While Word2Vec effectively captures word relationships and semantic meaning, the limited dataset size may have impacted its ability to generate high-quality embeddings, leading to reduced classification performance.

VII. DISCUSSION

This section provides insights into the observed model performance and elaborates on why each vectorization technique yielded particular results.

A. Why Count Vectorization Achieves the Highest Accuracy?

Count Vectorization performs well in terms of accuracy because it relies on simple term frequency, allowing it to capture common patterns in customer reviews efficiently. This method is highly effective when sentiment classification is primarily influenced by frequently used words. However, it does not account for word importance or contextual meaning, which may lead to over-reliance on frequent terms and potential bias toward majority classes.

B. Why TF-IDF is the Best Choice for Balanced Performance?

TF-IDF outperforms other methods in terms of macro-F1 score, demonstrating a better ability to differentiate between sentiment classes. Unlike Count Vectorization, which treats all words equally, TF-IDF assigns lower weights to common words and higher weights to distinguishing words, improving the model’s ability to recognize subtle variations in sentiment. This leads to more balanced performance across positive, neutral, and negative categories.

C. Why Word2Vec Underperformed?

While Word2Vec is powerful in capturing semantic relationships between words, its performance in this study was lower than expected. This could be attributed to several factors. First, small dataset size played a significant role, as Word2Vec requires large amounts of data to learn high-quality word embeddings. The limited number of Sephora reviews may have restricted its ability to effectively capture meaningful word relationships. Additionally, Word2Vec lacks contextual awareness, unlike advanced deep learning models such as transformers (e.g., BERT), which consider word position and sentence structure. This limitation makes it less effective for sentiment analysis, where context is crucial for correctly interpreting the tone and meaning of a review.

D. Choosing the Best Model Based on Priorities

The choice of vectorization method depends on the primary objective of the sentiment analysis. If accuracy is the main priority and the goal is to maximize correct classifications, Count Vectorization + Logistic Regression is the most effective option. However, if a balanced performance across sentiment categories is required—ensuring fair classification of positive, neutral, and negative reviews—TF-IDF + Logistic Regression proves to be the superior choice. This makes TF-IDF particularly valuable in cases where preventing bias toward the majority sentiment class is essential.

E. Implications and Future Research

The findings of this study demonstrate that TF-IDF is the most effective vectorization technique for sentiment classification in Sephora product reviews, offering a balanced approach between accuracy and class distribution fairness. However, there are several avenues for future research to further enhance sentiment classification performance. One potential improvement is the use of deep learning models such as BERT or LSTMs, which can better capture contextual meaning and sentiment nuances. Additionally, exploring hybrid models that combine TF-IDF with word embeddings could provide more comprehensive text representations, improving classification accuracy. Finally, expanding the dataset could help enhance the model's ability to generalize across a wider range of reviews, leading to a more robust sentiment classification system.

VIII. CONCLUSION

This study demonstrates the effectiveness of sentiment analysis in extracting valuable insights from Sephora product reviews, which in turn can guide strategic decisions for product development, marketing, and customer engagement in the cosmetics industry. By comparing three text vectorization techniques—Count Vectorization, TF-IDF, and Word2Vec—paired with logistic regression, we found that while Count Vectorization yields the highest overall accuracy, TF-IDF emerges as the most balanced approach, achieving a stronger macro-F1 score. TF-IDF's capacity to emphasize distinguishing words while downplaying frequent terms better addresses the imbalanced nature of the dataset and captures nuanced sentiment in customer feedback. Although Word2Vec holds promise due to its ability to capture semantic relationships, its reliance on large datasets and limited contextual information resulted in underperformance for this specific task.

These findings suggest that the choice of vectorization technique should be driven by the primary analytical objective: whether maximizing sheer accuracy or ensuring equitable representation across positive, neutral, and negative sentiments. From a practical standpoint, adopting TF-IDF combined with logistic regression enables more reliable detection of critical feedback and less frequent sentiments, assisting companies like Sephora in promptly identifying areas for improvement. Future research could extend these results by exploring more advanced deep learning approaches, such as BERT and LSTM models, which may further enhance the capture of contextual nuances. Moreover, collecting a larger and more diverse dataset—including additional product categories or consumer demographics—could improve the robustness of

sentiment classification. Ultimately, leveraging balanced, data-driven insights allows brands to elevate customer satisfaction, optimize product offerings, and maintain a competitive edge in the dynamic beauty market.

REFERENCES

- [1] Bing Liu. "Sentiment analysis and subjectivity". In: *Handbook of Natural Language Processing*. Ed. by N. Indurkha and F. J. Damerau. 2nd ed. 2010.
- [2] O. O. Hasas. "A study of sentiment analysis on customer reviews". Master's thesis. University of Missouri-Columbia, 2023.
- [3] R. Prabowo and M. Thelwall. "Sentiment analysis: A combined approach". In: *Journal of Informetrics* 3.2 (2009), pp. 143–157.